

Using Python Libraries from Tests

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1 Introduction

Python is a winner in the trend market of programming languages in this moment. Although the syntax is so comfortable to the programmers, the clue is in the modern libraries for system administrators, scientifics and with low importance for web programmers.

You can see the Tiobe Index:

Jan 2021	Jan 2020	Change	Programming Language	Ratings	Change
1	2	▲	C	17.38%	+1.61%
2	1	▼	Java	11.96%	-4.93%
3	3		Python	11.72%	+2.01%
4	4		C++	7.56%	+1.99%
5	5		C#	3.95%	-1.40%
6	6		Visual Basic	3.84%	-1.44%
7	7		JavaScript	2.20%	-0.25%
8	8		PHP	1.99%	-0.41%
9	18	▲	R	1.90%	+1.10%
10	23	▲	Groovy	1.84%	+1.23%
11	15	▲	Assembly language	1.64%	+0.76%
12	10	▼	SQL	1.61%	+0.10%
13	9	▼	Swift	1.43%	-0.36%
14	14		Go	1.41%	+0.51%

You can check that Python is growing fast and reaching the most important positions in the market. The third position in January of 2021.

You can use or don't use a test system such as nose or pytest, but all people need test the code in the programming. After to write software you must introduce inputs in your code and verify if the output is ok.

The advantage using a test system is generate automation in the process about inputs, code being tested and outputs. Each test is evaluated with asserts, that's comparisons between the result that you deserve and the result obtained. For instance,

```
def test_models_lda(self):
    X = np.array([[-1, -1], [-2, -1], [-3, -2],
                  [1, 1], [2, 1], [3, 2]])
    y = np.array([1, 1, 1, 2, 2, 2])
    clf = LinearDiscriminantAnalysis()
    clf.fit(X, y)
    self.assertEqual(clf.predict([[-1, -1]]), 1)
```

So, assertEquals is the comparison and the rest is the code to be verified.

To become a professional in Python you must learn a good bunch of libraries. So you can practice the libraries reading tutorials, python notebooks, or you could write tests about this libraries and perhaps to share these tests in some software repository.

If you save all test code about your python libraries in one or various software repositories you can test all code with a command. The python libraries can suffer chances, but you have the control about the use that you do about your libraries.

The goal of this book is to teach Python Libraries from the tests.

2 Installation

You can create a python virtual environment and to install damelibraries with python package:

```
$ python3 -m venv /tmp/venv
$ cd /tmp/venv
$ source bin/activate
$ pip install --upgrade pip
$ pip3 install damelibraries
$ cd lib/python3.5/site-packages/damelibraries
```

Now you can execute tests to check different libraries:

```
$ cd /tmp/venv/lib/python3.8/site-packages/damealgorithms
$ nosetests3 tests
```

```
$ cd /tmp/venv/lib/python3.8/site-packages/damejson
$ nosetests3 tests
```

```
$ cd /tmp/venv/lib/python3.8/site-packages/damenltk
$ nosetests3 tests
```

```
$ cd /tmp/venv/lib/python3.8/site-packages/damenumpy
$ nosetests3 tests
```

```
$ cd /tmp/venv/lib/python3.8/site-packages/damepandas
$ nosetests3 tests
```

```
$ cd /tmp/venv/lib/python3.8/site-packages/damescikit
$ nosetests3 tests
```

```
$ cd /tmp/venv/lib/python3.8/site-packages/damescipy
$ nosetests3 tests
```

3 DameNLTK

NLTK is an acronym about Natural Language ToolKit.

A summary of tasks about Natural Language Processing could be:

- Sentiment Analysis
- Detect Gender
- Sentence Similarity
- Text Summary
- Classify Documents
- Manage Words
- Gramatical Trees
- Extract Keywords
- Disambiguation

You could be in the path of nltk so:

```
$ cd /tmp/venv/
$ source bin/activate
(venv)$ pip3 install damenltk
$ cd /tmp/venv/lib/python3.8/site-packages/damenltk
$ nosetest3 test
```

3.1 Sentiment Analysis

Sentiment analysis is applied in a popular way in Twitter with python libraries such as tweepy. Finally, it is a classification task about words with a label (the sentiment). That could be implemented with nltk. Take a look to the test:

```
import unittest
from nltk.classify import SklearnClassifier
class TddInPythonExample(unittest.TestCase):

    def test_sentiment_variable_create_returns_correct_result(self):
        pos_tweets = [('I love this car', 'positive'),
                       ('This view is amazing', 'positive'),
                       ('I feel great this morning', 'positive'),
                       ('I am so excited about the concert', 'positive'),
                       ('He is my best friend', 'positive'),
                       ('This movie was great', 'positive'),
                       ('This movie was not pathetic', 'positive')]

        neg_tweets = [('I do not like this car', 'negative'),
                       ('This view is horrible', 'negative'),
                       ('I feel tired this morning', 'negative'),
                       ('I am not looking forward to the concert', 'negative'),
                       ('He is my enemy', 'negative'),
                       ('This is a pathetic movie', 'negative')]
```

```

tweets_with_sentiment = []
for (tweet, sentiment) in pos_tweets + neg_tweets:
    filtered_tweet_words = [word.lower() for word in tweet.split() if len(word) > 3]
    tweets_with_sentiment.append((filtered_tweet_words, sentiment))

self.assertEqual([(['love', 'this', 'car'], 'positive'),
                   ([ 'this', 'view', 'amazing'], 'positive'),
                   ([ 'feel', 'great', 'this', 'morning'], 'positive'),
                   ([ 'excited', 'about', 'the', 'concert'], 'positive'),
                   ([ 'best', 'friend'], 'positive'),
                   ([ 'this', 'movie', 'was', 'great'], 'positive'),
                   ([ 'this', 'movie', 'was', 'not', 'pathetic'], 'positive'),
                   ([ 'not', 'like', 'this', 'car'], 'negative'),
                   ([ 'this', 'view', 'horrible'], 'negative'),
                   ([ 'feel', 'tired', 'this', 'morning'], 'negative'),
                   ([ 'not', 'looking', 'forward', 'the', 'concert'], 'negative'),
                   ([ 'enemy'], 'negative'),
                   ([ 'this', 'pathetic', 'movie'], 'negative')],
                  tweets_with_sentiment)

```

You can execute this test with:

```
$ nosetest3 test/test_sentiment.py
```

3.2 Detect Gender

The gender detection from the name is a NLP task. So the input is one or more word (name and perhaps surname) and the output is a gender male, female, ...

NLTK brings a good list of names in the `nlk.corpus`. So you can classify names with two ways:

- Querying single files
- Supervised classifier

The python method using single files would be:

```

def gender_name(self, name):
    gender = 'undefined'
    if name in names.words('male.txt'):
        gender = 'male'
    if name in names.words('female.txt'):
        if (gender == 'male'):
            gender = 'both'
        else:
            gender = 'female'
    return gender

```

And the python test would be:

```

def test_gender_name_method_returns_correct_result(self):
    dn = DameNLTK()

```



```

self.assertTrue(dn.gender_name("Neo"), "male")
self.assertTrue(dn.gender_name("Trinity"), "female")
self.assertTrue(dn.gender_name("Andrea"), "both")

```

The python methods using machine learnig would be:

```

def gender_features(self, word):
    return 'last_letter': word[-1]

def gender_classifier(self):
    labeled_names = [(name, 'male') for name in names.words('male.txt')] +
                    [(name, 'female') for name in names.words('female.txt')]

    featuresets = [(self.gender_features(n), gender) for (n, gender) in labeled_names]
    train_set, test_set = featuresets[500:], featuresets[:500]
    classifier = nltk.NaiveBayesClassifier.train(train_set)
    return classifier

```

And the python tests would be:

```

def test_gender_classifier_method_returns_correct_result(self):
    dn = DameNLTK()
    classifier = dn.gender_classifier()
    self.assertTrue(classifier.classify(dn.gender_features("Neo")), "male")
    self.assertTrue(classifier.classify(dn.gender_features("Trinity")), "female")

```

You can use an advanced software to detect gender such as Damegender with many features:

```
$ pip3 install damegender[all]
```

You can learn to use the software with tests oriented to the user (bash files) or tests oriented to the developer (python tests).

```

$ python3 -m venv /tmp/damegender/
$ cd /tmp/damegender/
$ source bin/activate
$ pip3 install damegender[all]
$ ./testsbycommands.sh
maindavid test is ok
mainjesus test is ok
mainines test is ok
mainalex test is ok
mainandrea test is ok
mainjesusmaria test is ok
mainjosemaria test is ok
mainelenaluciahelena test is ok
mainjuliaus test is ok
mainjuliauk test is ok
mainjuliauy test is ok
[...]

$ nosetest3 test

```

[...]

Execute these tests could be long time, but you can execute a single file test, for example, you can execute `test/test_dame_utils.py`:

```
$ nosetest3 test/test_dame_utils.py
```

Or you can execute a single test in a file with:

```
$ nosetests3 test/test_dame_utils.py:TddInPythonExample.test_string2array■
```

3.3 Sentence Similarity

Appendix A License

Version 1.3, 3 November 2008

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